Assignment I : Portfolio Optimization

Objective

Students must design, backtest, and optimize a **multi-asset portfolio** using a 10-stock universe selected from the provided 3-year daily returns dataset. The goal is to apply **Modern Portfolio Theory (MPT)** to engineer a weighting strategy that maximizes risk-adjusted performance.

1. Dataset Provided:

The dataset consists of two files one for the name of the stocks and their sectors, the other one is the daily returns of these stocks for the last 3 years. There are a total of 30 stocks.

2. Technical Constraints:
   - **Asset Universe:** n = 10 (Selection must be justified via correlation analysis).
   - **Long-Only:** $w_i >= 0$ (No short selling).
   - **Budget:** $\sum_{i=1}^{n} W$
   - **Risk-Free Rate (R_f):** 6.5%.

3. Tools you can use:

    Excel, Python or any coding language.

4. Deliverables
   a. **Work**
      i. Excel File or Python Notebook file
   b. **Report:** ***1-pager pdf***
      i. **Optimization Plot:** A visual Efficient Frontier generated via Monte Carlo simulation (minimum 10,000 iterations).
      ii. **Correlation Matrix:** A heatmap displaying the linear dependency between the 10 selected assets.
      iii. **Weight Attribution:** A table comparing the weights of the **Global Minimum Variance (GMV)** vs. the **Maximum Sharpe** portfolios.